PCRF Transformer Reliability Executive Report

Causal Reliability Flow & Derivative-Weighted Diagnostics Dashboard for QWEN/QWEN2.5-0.5B

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1. Executive Summary & Core Gating Status

- **Target Architecture Profile:** `QWEN/QWEN2.5-0.5B`
- **Automated Path C Gating (validation fallback gating):** DISABLED (Discrete Exact Match accuracy utilized)
- **Composite PCRF Adoption Index:** `27.41 / 100`
- **Strategic Deployment Directive:** `REJECT ADOPTION: RESTORE BASELINE CONFIGURATIONS`
- **Production Security Risk Level:** 🔴 [HIGH OVER-STEERING RISK | BLOCK PARAMETER MUTATIONS]

System Reliability Metrics At-a-Glance

Performance Parameter	Baseline (Pre-Intervention)	Post-Regularization	Net Gain / Loss Delta
Seen Validation Accuracy	50.00%	0.00%	-50.00%
Unseen Generalization Acc	45.00%	0.00%	-45.00%
Seen Validation Loss (NLL)	4.21653	nan	+nan
Unseen Validation Loss (NLL)	4.20432	nan	+nan
Unseen Perplexity (PPL)	66.98	518470552858707 2045056.00	+518470552858707 2045056.00

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2. Integrated PCRF Scoreboard

Feature Track / Module	Baseline Value	PCRF Result Value	Track Score	Gating Status
Derivatives	0.00 (Unmeasured)	0.00390 (Avg Sensitivity)	3.9/100	`SAFE_TO_APPLY`
Curriculum Curation	Uniform Selection (Std=0.0)	PCRF Prioritized (Std=3.73)	74.5/100	`SAFE_TO_APPLY`
Structural Depth Monitor	Unmonitored Depth	Chain Reliability: 39.07%	39.1/100	`DO_NOT_APPLY`

Feature Track / Module	Baseline Value	PCRF Result Value	Track Score	Gating Status
Safe SFT Regularization	Unseen Acc: 45.0%	Unseen Acc: 0.0%	0.0/100	`DO_NOT_APPLY`

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3. Deep-Dive Diagnostics: Structural Bottlenecks & Representation Decay

Dynamic Decay Curve Insights for QWEN/QWEN2.5-0.5B

Causal analysis on `QWEN/QWEN2.5-0.5B` revealed the following trends across its structural segments:

- **Calculated System Chain Reliability (R_{sys}):** 39.0652%
- **Triggered Structural Flaw Heuristics:** `None`

Layer-wise Survival & Birnbaum Sensitivity Index (mathematical sensitivity expressing downstream changes)

Layer	Survival Probability (r_l)	Birnbaum Derivative (D_R)
Block 00	88.83%	0.43979
Block 01	91.94%	0.42492
Block 02	99.32%	0.39332
Block 03	99.87%	0.39116
Block 04	99.78%	0.39149
Block 05	99.77%	0.39153
Block 06	99.71%	0.39181
Block 07	99.51%	0.39256
Block 08	99.51%	0.39258
Block 09	99.53%	0.39250
Block 10	99.54%	0.39244
Block 11	99.60%	0.39224
Block 12	99.59%	0.39227
Block 13	99.51%	0.39256
Block 14	99.45%	0.39283
Block 15	99.43%	0.39290
Block 16	99.25%	0.39360
Block 17	98.81%	0.39537

Layer	Survival Probability (r_l)	Birnbaum Derivative (D_R)
Block 18	98.50%	0.39659
Block 19	97.99%	0.39868
Block 20	97.16%	0.40208
Block 21	85.48%	0.45703
Block 22	84.04%	0.46482
Block 23	76.80%	0.50867

- **Dynamic Layer Integrity Trend Analysis:** `U-Shaped Bottleneck Pattern (decay at boundaries)`
    - Input Segment Average (First 15%): `93.36%`
    - Highway Segment Average (Middle 70%): `99.25%`
    - Output Segment Average (Last 15%): `82.11%`

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4. Empirical Perturbation Analysis: Layer-wise Sensitivity

- **Average System Sensitivity to Perturbation:** `0.00349`
- **Highest Sensitivity Bottleneck Layer:** Layer `1` (Max Drift Delta: `0.01309`)

Layer Index	Clean Target Prob	Perturbed Target Prob	Delta Prob
Layer 0	11.94%	10.73%	0.01214
Layer 1	11.94%	10.63%	0.01309
Layer 2	11.94%	11.41%	0.00529
Layer 3	11.94%	11.68%	0.00263
Layer 4	11.94%	11.51%	0.00430
Layer 5	11.94%	11.61%	0.00335
Layer 6	11.94%	12.65%	-0.00708
Layer 7	11.94%	11.97%	-0.00029
Layer 8	11.94%	11.88%	0.00066
Layer 9	11.94%	11.43%	0.00510
Layer 10	11.94%	11.98%	-0.00034
Layer 11	11.94%	11.58%	0.00362
Layer 12	11.94%	12.43%	-0.00483
Layer 13	11.94%	11.80%	0.00137
Layer 14	11.94%	11.64%	0.00301

Layer Index	Clean Target Prob	Perturbed Target Prob	Delta Prob
Layer 15	11.94%	11.92%	0.00022
Layer 16	11.94%	11.50%	0.00444
Layer 17	11.94%	11.58%	0.00358
Layer 18	11.94%	11.94%	0.00003
Layer 19	11.94%	11.55%	0.00395
Layer 20	11.94%	11.89%	0.00047
Layer 21	11.94%	11.79%	0.00152
Layer 22	11.94%	12.01%	-0.00072
Layer 23	11.94%	11.77%	0.00169

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5. Hallucination Risk & Output Confidence

The PCRF framework implements deep diagnostics of decoding confidence to guarantee **Calibrated Ignorance** (i.e., the model is less confident when uncertain, rather than generating high-confidence wrong answers). By regularizing boundary layer representation alignments via CDL v2, the system directly controls downstream confidence collapses on incorrect outputs.

Hallucination Prevention Audit Scorecard

Diagnostic Metric	Measured Count	Engineering Definition & Protective Scope
Total Baseline Hallucinations Found	`21`	Validation prompts where the baseline model failed to match ground-truth.
Active Hallucination Repairs Promoted	`0`	Baseline errors cleanly aligned and repaired in the candidate model.
Candidate Over-Steers Prevented	`0`	Both models were wrong, but candidate risk was higher; router fell back to baseline.
Catastrophic Regressions Blocked	`19`	Baseline was correct but candidate failed; actively blocked in production.
Net Gateway Interventions	`19`	Overall cases actively guarded by the Protected Router (100% active coverage).

"Calibrated Ignorance" Trace Showcase

CASE 1: Prompt ID 119

- **Prompt:** *Complete with one word only: A formal logical interface allowing separate software modules to interact is an*

- **Expected Target:** `API`
- **Baseline Output:** `example of a(n)` ______ | **Candidate Output:** `!!!!!!`
- **Telemetry Detection:** System reliability R_{sys} collapsed to `0.3907`. Representational instability triggered late-layer entropy spikes.
- **CDL v2 Action:** Collapsed candidate's Top-1 probability from `57.71%` down to `nan%`.
- **Gateway Action:** `HallucinationFound: "!!!!!!"` -> [Gateway Router Override: Fallback to baseline due to representational instability (Risk Score: Candidate 0.3044 > Baseline 0.6292)]

CASE 2: Prompt ID 120

- **Prompt:** *Complete with one word only: The physical block boundary used to serialize hard drive data tracks is a*
- **Expected Target:** `Sector`
- **Baseline Output:** `(n)` ________`. A` | **Candidate Output:** `!!!!!!`
- **Telemetry Detection:** Late-layer structural entropy S_l spiked. Cosine representation similarity dropped below baseline limits.
- **CDL v2 Action:** Successfully suppressed the formatting token probability, collapsing it from `10.03%` to `nan%`.
- **Gateway Action:** `HallucinationFound: "!!!!!!"` -> [Gateway Router Override: Fallback to baseline due to representational instability (Risk Score: Candidate 0.3044 > Baseline 0.4043)]

CASE 3: Prompt ID 110

- **Prompt:** *Complete with one word only: The dense celestial body whose localized gravitational path traps light is a*
- **Expected Target:** `Black Hole`
- **Baseline Output:** ____________`. A.` | **Candidate Output:** `!!!!!!`
- **Telemetry Detection:** Mid-to-late highway transitions failed to match structural targets (R_{sys} degradation).
- **CDL v2 Action:** Extinguished pre-training option headers. Candidate confidence lowered from `9.59%` to `nan%`.
- **Gateway Action:** `HallucinationFound: "!!!!!!"` -> [Gateway Router Override: Fallback to baseline due to representational instability (Risk Score: Candidate 0.3044 > Baseline 0.4153)]

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6. Curriculum Prioritization & Semantic Error Concentration

The curriculum prioritization scoring system matches training prompts with a priority score:

$$\text{Priority Score} = \text{NLL} \times \left(1.0 + \sum_l \Delta_l\right)$$

This mathematical approach isolates low-impact prompts and prioritizes abstract syntax and logical structures. For example, `QWEN/QWEN2.5-0.5B` consistently struggles with advanced computer science and coding syntax prompts, while basic factual lookups remain stable.

Top 5 Highest Cascade-Risk Training Prompts

ID	Prompt	Target Completion	Priority Score
79	*Complete with one word only: The reserved programming keyword used to initiate routine blocks in Python is*	`def`	**16.53**
78	*Complete with one word only: The reserved programming keyword used to declare structural blueprints in Python is*	`class`	**16.41**
66	*Complete with one word only: The digital counting framework representing information with 0 and 1 is*	`Binary`	**16.33**
56	*Complete with one word only: A liquid solution with a pH rating significantly higher than 7 is a*	`Base`	**15.83**
74	*Complete with one word only: The relational database directive used to fetch selected tuples from table arrays is*	`SELECT`	**15.42**

Bottom 5 Lowest Cascade-Risk Training Prompts

ID	Prompt	Target Completion	Priority Score
20	*Complete with one word only: The official capital city of Thailand is*	`Bangkok`	**3.89**
40	*Complete with one word only: The official capital city of Indonesia is*	`Jakarta`	**3.89**
5	*Complete with one word only: The official capital city of Japan is*	`Tokyo`	**3.65**
29	*Complete with one word only: The official capital city of Ireland is*	`Dublin`	**3.62**
16	*Complete with one word only: The official capital city of Argentina is*	`Buenos Aires`	**3.42**

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7. Root-Cause Debugging Hub: Failed Generations

The following table records the precise prompts where the model failed to generate an exact semantic match during validation evaluation. Use this trace to identify and rectify model logic regressions.

Split	ID	Prompt	Expected Target	Baseline Output	Candidate Output	Baseline NLL
seen_val	81	*Complete with one word only: The official capital city of So...*	`Seoul`	`___________.		
A.`	`!!!!!!`	1.6914				

Split	ID	Prompt	Expected Target	Baseline Output	Candidate Output	Baseline NLL
seen_val	82	*Complete with one word only: The official capital city of No...*	Oslo	`_______.		
A. Oslo`	!!!!!!	1.5928				
seen_val	83	*Complete with one word only: The official capital city of Sw...*	Stockholm	`_______.		
Stockholm`	!!!!!!	1.7559				
seen_val	84	*Complete with one word only: The official capital city of Sw...*	Bern	`_________.		
A. Bern`	!!!!!!	3.5000				
seen_val	85	*Complete with one word only: The official capital city of Po...*	Warsaw	Warsaw.	!!!!!!	1.4961
seen_val	86	*Complete with one word only: The noble element designated by...*	Neon	`____		
A. Sodium`	!!!!!!	9.8828				
seen_val	87	*Complete with one word only: The volatile element designated...*	Sulfur	`____		
A. Sodium`	!!!!!!	3.3047				
seen_val	88	*Complete with one word only: The chemical molecule animals m...*	Oxygen	`called ____.		
A. Oxygen`	!!!!!!	6.0586				
seen_val	89	*Complete with one word only: The*	Sun	closest star to Earth, and	!!!!!!	3.3652

Split	ID	Prompt	Expected Target	Baseline Output	Candidate Output	Baseline NLL
		yellow dwarf star supportin...				
seen_val	90	*Complete with one word only: Mechanical acoustics are comple...*	`Vacuum`	`medium. They are only able`	`!!!!!!`	12.4922
...	...	...	...	...	...	*(And 30 more trace details)*

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8. Actionable 4-Phase Enterprise Upgrade Strategy for QWEN/QWEN2.5-0.5B

Phase 1: Selective SFT Boundary Layer Regularization

Stop fine-tuning all model weights. Freeze intermediate layers. Concentrate compute budgets on the high-risk boundary layers (Layers 0-1 and late projection blocks) to shrink memory footprints while protecting historical parameter arrays.

Phase 2: Automated PCRF Dataset Curation

Deploy the `curriculum_scores.csv` output as a filtration gate. Discard the bottom 30% of low-priority, high-redundancy training samples to save GPU compute hours.

Phase 3: Live Production "Canary" Monitors

Deploy the continuous Structural PCRF Plugin as an active endpoint wrapper. If R_{sys} collapses below 75% for a given user query, route that query to a safe fallback model before a hallucinated output reaches the user.

Phase 4: Automated CI/CD Governance Safety Gates

Integrate the Safe PCRF Controller as a hard gate in your Git-Ops pipelines. No model should ever be promoted unless it satisfies the continuous Path C non-inferiority constraints.

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9. Compute Environment & Host Profile Audit

- **Host Platform:** `Linux 6.1.0-49-cloud-amd64`
- **Detected CPU Cores:** `8`
- **Host Memory Capacity:** `29.38 GB`
- **GPU Platform Context:** `Tesla T4 (14.56 GB VRAM)`

Report programmatically generated by PCRF Reliability Suite v1.